Alpha decay 3

In this section we derive a regression formula from the data set

	$\log \tau_{1/2}$	E
^{226}U	-1.31	7.70
^{228}U	6.30	6.80
^{230}U	14.4	5.99
^{232}U	21.5	5.41
^{234}U	29.7	4.86
^{236}U	34.2	4.57
^{238}U	39.5	4.27

From the previous sections we have

$$\log \tau_{1/2} \propto \gamma$$

and

$$\gamma = C_1 \frac{Z}{\sqrt{E}} - C_2 \sqrt{Z r_1}$$

Combine the formulas as

$$\log \tau_{1/2} = \beta_1 \frac{90}{\sqrt{E}} + \beta_0$$

The nuclear radius $r_1 \propto A^{1/3}$ is discarded because it does not change very much.

By ordinary least squares we obtain

$$\beta_1 = 3.682$$
$$\beta_0 = -120.8$$

Hence

$$\log \tau_{1/2} = 3.682 \frac{90}{\sqrt{E}} - 120.8$$

From the regression formula we obtain the following predicted values.

$\log \tau_{1/2}$	Predicted $\log \tau_{1/2}$
-1.31	-1.40
6.30	6.27
14.4	14.6
21.5	21.6
29.7	29.5
34.2	34.1
39.5	39.5

The observed linearity of $\log \tau_{1/2}$ and $Z/\sqrt{E}$ confirms the tunneling hypothesis for alpha decay.

Eigenmath script